

UTS-UAV: A Context-Grounded Benchmark for Operational Event Discovery in Low-Altitude UAV Imagery

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Abstract—Operational UAV inspection requires more than classifying a supplied crop: a system must reject ordinary frames, find a region that contains sufficient evidence, and recognize the event under severe scale and deployment shifts. We introduce UTS-UAV, a real low-altitude UAV benchmark with three defining properties: (1) *operational realism*, pairing event-bearing frames with true no-event negatives from the same acquisition process; (2) *context-grounded supervision*, linking each positive full frame to a human-accepted contextual ROI, its high-resolution crop, and an event label; and (3) *deployment-oriented evaluation*, with leakage-controlled session-disjoint, unseen-site, and forward-temporal protocols. Five controlled tasks isolate event presence, contextual grounding, oracle-region recognition, joint discovery, and cascaded deployment. We benchmark closed-set detectors, open-vocabulary grounders, vision foundation models, multimodal large language models, and modular cascades, and propose Localize–Inspect–Recognize (LIR), a unified coarse-to-fine baseline with a learned no-event state, adaptive tiling, high-resolution ROI reinspection, global–local semantic fusion, and a counterfactual evidence verifier, and a residual-risk controller that allocates additional views only when needed. The evaluation further covers calibration, hard negatives, long-tail classes, resolution, prompt/output format, evidence sufficiency, matched-compute efficiency, adaptive call budgets, corruptions, aerial pretraining, and geographic/temporal transfer. *Result placeholder: Insert the best end-to-end gain, negative-frame false-positive reduction, budget–accuracy AUC gain, evidence-sufficiency gain, and unseen-site retention here.*

I. INTRODUCTION

Low-altitude UAVs can survey farmland, waterways, construction zones, and settlements at a scale that is impractical for ground patrols. A deployable inspection system must answer three coupled questions: *is* an event present, *where* is the evidence-bearing region, and *what* event does it represent? The first question is easily hidden by positive-only datasets; the second is difficult because useful evidence may occupy a tiny fraction of a 4K frame; and the third is contextual because labels such as exposed waste, unclean water, or active construction describe conditions, not pixel-tight object categories.

Existing aerial datasets are rich but leave this combination under-specified. DOTA and VisDrone emphasize generic objects under scale and viewpoint variation [1], [2]; ERA and CapERA study event recognition or description from selected clips [3], [4]. Recent UAV vision–language benchmarks greatly broaden semantic, spatial, and planning evaluation [5]–[7], while AirZoo contributes geometry-rich supervision and cross-domain transfer tracks [8]. These advances do not directly quantify the failure chain from ordinary-scene rejection to context-region discovery and fine-grained operational recognition. A crop classifier can therefore appear deployment-

ready without ever searching a full frame or facing a true negative.

The closest recent grounding benchmarks sharpen this gap. AerialVG and DVGBench test explicit or implicit references in high-resolution drone imagery [9], [10]; UniRef-UAV additionally admits absent targets and variable output cardinality [11]. Their targets remain referred objects. In operational inspection, by contrast, the query is implicit (“find a reportable event”), the positive region may be a diffuse condition, and the first correct output for many frames is no-event.

We build UTS-UAV from a real operational archive while treating provenance as part of the benchmark contract. Full-frame/crop pairs were linked, all contextual boxes were recovered and manually accepted, source labels were frozen, and genuine no-event frames were retained. Content, session, and site groups are assigned before splitting, and the test labels are sealed. The benchmark consequently supports controlled counterfactuals: a crop estimates the recognition ceiling; a ground-truth ROI tests whether full-frame compression and background are harmful; presence and class-agnostic grounding measure search; and joint or cascaded inference measures end-to-end readiness.

Following the property–track organization of large dataset papers and the capability–diagnostic structure of recent UAV benchmarks, we make four contributions:

- **A context-grounded operational benchmark.** UTS-UAV combines true positive and no-event frames with full-resolution imagery, contextual ROIs, crops, event labels, and leakage-safe acquisition groups.
- **Five interoperable tasks and three shifts.** A single record contract supports presence-aware grounding, two oracle recognition settings, unified discovery, and cascaded discovery under session, site, and temporal generalization.
- **A broad, reproducible model study.** We compare detectors, open-vocabulary grounders, visual encoders, generative VLMs, and cascades using presence, localization, classification, calibration, efficiency, and end-to-end metrics with group-bootstrap uncertainty.
- **A unified coarse-to-fine baseline and diagnostic suite.** LIR learns an explicit no-event state and contextual region proposal, reinspects source pixels, fuses local, global, and text evidence, and verifies a region through evidence-preserving/evidence-removing counterfactuals. A residual-risk controller adaptively spends visual calls. Fourteen analyses separate resolution, localization, semantics, output validity, evidence sufficiency, long-tail

Operational event discovery: reject, ground, and recognize

Positive frames contain a contextual evidence ROI; genuine no-event frames require an empty prediction.

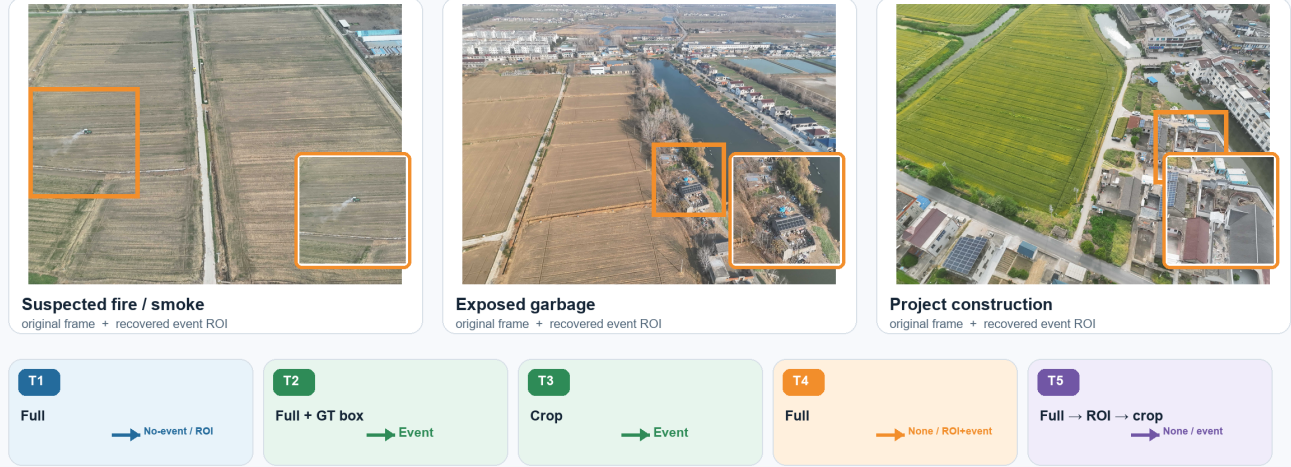


Fig. 1. UTS-UAV evaluates full-frame operational event discovery. Positive examples (shown) link a full frame, a human-accepted contextual ROI, and a crop; no-event frames have an empty ROI target. The five tasks isolate presence/localization, oracle-region recognition, joint prediction, and a modular cascade. Boxes denote sufficient contextual evidence rather than pixel-tight physical objects.

behavior, negative-frame calibration, budget allocation, corruption robustness, public aerial transfer, and shift failure.

Crucially, our box is an *evidence ROI*. It is evaluated as a contextual grounding target and must not be interpreted as a segmentation-derived or pixel-tight object-detection box.

II. RELATED WORK

Aerial perception and operational events. DOTA and VisDrone established aerial detection under extreme scale, orientation, and density variation [1], [2]; UAV-Human and MAVREC extend aerial understanding to human behavior and unscripted multi-view recognition [12], [13]. ERA and CapERA target aerial event recognition and captioning [3], [4], whereas A2Seek studies reasoning-centric video anomaly localization [14]. UTS-UAV focuses on static, high-resolution operational inspection and explicitly couples true no-event rejection, contextual region grounding, and event recognition.

UAV foundation-model benchmarks. GeoChat introduced grounded region interaction for remote sensing [15]. UAVBench/UAVID-1M evaluates ten image- and region-level tasks and provides large-scale instruction tuning [5]; MM-UAVBench organizes 19 subtasks around perception, cognition, and planning [6]; and SpatialUAV uses 14 task types, seven input configurations, and nine answer formats to diagnose low-altitude spatial intelligence [7]. SpatialSky-Bench further exposes navigation-specific geometric weaknesses [16]. AirZoo organizes aerial retrieval, cross-view matching, and reconstruction around large-scale geometric pretraining and real-domain transfer [8]. Inspired by their systematic taxonomies, we organize one operational decision into five causally linked tasks and ask where the discovery chain fails.

Newest UAV suites further emphasize grounded, inspectable answers. UAVReason unifies aerial reasoning and dense generation [17]; UAV-DualCog requires spatial or temporal grounding and reports a human reference [18]; and UAVQA-Bench evaluates tool-routed training-free agents with intermediate verification and difficulty-adaptive search [19]. We adopt this budget-aware diagnostic spirit for a static operational chain with sealed no-event/ROI/event outputs.

Grounding, detection, and unified vision models. Grounding DINO grounds arbitrary text in boxes [20], YOLO-World provides efficient open-vocabulary detection [21], and D-FINE improves real-time DETR localization [22]. Florence-2 expresses detection and grounding in a prompt-driven sequence representation [23]; RegionGPT and Qwen3-VL strengthen region and coordinate reasoning [24], [25]; DINOv2 supplies transferable visual features [26]. AerialVG exposes the high-resolution and relation-reasoning bottleneck [9], DVGBench targets implicit references [10], and UniRef-UAV explicitly measures no-target discrimination [11]. Operational events differ from canonical objects: their regions can be diffuse and their names need not describe a single visible entity. We therefore compare closed-set, open-vocabulary, and generative formulations, and introduce a context-aware no-event model rather than assuming that every frame contains a target.

Leakage, calibration, and deployment shift. Adjacent UAV frames and repeated site visits are highly correlated. Random image splits can reward memorization of place, weather, or capture trajectory. Our content-, session-, and site-grouped protocols, forward-time test, sealed labels, and session-level bootstrap make generalization and uncertainty part of the benchmark definition. Negative-frame calibration is evaluated jointly with localization because a useful inspection

system must trade recall against false alarms rather than optimize positive-only accuracy. Aerial Mirage documents hallucination in aerial LVLM outputs [27]; accordingly, invalid strings, unsupported event claims, and plausible boxes on negatives remain scored failures in our evaluator.

III. UTS-UAV BENCHMARK

A. Three Benchmark Properties

Operational realism. UTS-UAV contains **TBD** full-resolution frames, comprising **TBD** event-bearing samples, **TBD** true no-event samples, and **TBD** operational event classes. Negatives are captured by the same operational process rather than synthesized from random background patches, enabling meaningful false-alarm and calibration analysis.

Context-grounded supervision. A sample is

$$d_i = (I_i, p_i, b_i, C_i, y_i, m_i), \quad (1)$$

where I_i is the full frame, $p_i \in \{0, 1\}$ is event presence, b_i is a contextual evidence ROI, C_i is its high-resolution crop, y_i is the event label, and m_i contains opaque acquisition groups. For $p_i = 0$, $(b_i, C_i, y_i) = \emptyset$; for $p_i = 1$, the full/ROI/crop views share one record identity. The ROI encloses evidence sufficient for an operator to judge the event. It can include surrounding water, terrain, buildings, or activity context and is not a pixel-tight physical-object boundary.

Deployment-oriented protocols. Content, acquisition session, site, and time groups are frozen before model development. This supports controlled evaluation on repeated operations, unseen geography, and future capture periods without allowing near-duplicate frames to cross a boundary.

B. Construction and Quality Assurance

Positive full/crop pairs are linked using delivery keys and media hashes. Crop-to-frame registration proposes each ROI using local features, geometric verification, and corner projection; every proposed box is subsequently accepted or corrected by human review. Negatives are verified as containing no benchmark event under the frozen ontology. Label adjudication retains the source operational distinctions, with a supported-class macro average and a separate rare-class report. Company authorization, privacy review, release redaction, and test-label sealing are completed. Public records expose only opaque session/site identifiers and the metadata necessary to reproduce the protocols.

C. Five Controlled Tasks

T1: presence-aware contextual grounding maps a full frame to either no-event or one ROI. **T2: oracle-box recognition** supplies a positive full frame and its GT ROI as drawn, coordinate, or masked-background prompt. **T3: oracle-crop recognition** classifies the delivered positive crop and estimates the recognition ceiling. **T4: unified discovery** returns no-event or an ROI–event pair in one call. **T5: modular discovery** first proposes no-event/ROI and then classifies a source-resolution crop. T2–T3 isolate

representation and context; T1 isolates presence/search; T4–T5 compare unified and modular deployment while sharing the same records.

D. Leakage-Controlled Protocols

The primary **session-disjoint** protocol assigns entire acquisition and content groups to train, validation, or test (**TBD/TBD/TBD** frames). **Unseen-site** holds opaque geographic groups out for test (**TBD/TBD/TBD**), and **forward-temporal** trains on past captures and tests on later months (**TBD/TBD/TBD**). Positive and negative records obey the same grouping constraints. Every derived task keeps the parent record assignment, and all model, threshold, prompt, and context margin selection is validation-only.

E. Metrics and Statistical Protocol

Presence reports AP, AUROC, F1, and negative-frame false-positive rate (N-FPR) at a validation-selected operating point; calibration reports ECE and Brier score. On positive frames, contextual localization reports AP and recall at IoU $\{0.25, 0.50, 0.75\}$, mean IoU, and normalized center error. Lower IoU thresholds are retained because the annotation is a sufficient-context region, not a physical-object extent. Recognition reports supported-class macro-F1, accuracy, and tail macro-F1.

End-to-end contextual grounding AP (G-mAP@ τ) treats a prediction as a match only when its event label is correct and $\text{IoU}(\hat{b}, b) \geq \tau$; predictions on no-event frames are false positives. We also report positive-frame joint success $J@ \tau$, valid-output rate, parameters, peak memory, throughput, and median/P90 latency. All confidence intervals use paired bootstrap over acquisition sessions. Slices are fixed before test evaluation: ROI-area quartile, class frequency, boundary clarity, time period, site, and hard-negative subtype.

Because IoU alone cannot establish that a contextual box carries the decision evidence, we add a frozen *evidence-sufficiency* diagnostic. A committee of T3 recognition probes measures macro-F1 when given only the proposed ROI (F_{in}) and after that ROI is removed from the full frame (F_{out}). The evidence-sufficiency gap $\text{ESG} = F_{\text{in}} - F_{\text{out}}$ is reported with each grounding system, but is not folded into G-mAP. This counterfactual test rewards regions that are both sufficient and consequential without pretending their edges are object boundaries.

IV. LOCALIZE–INSPECT–RECOGNIZE

Generic VLMs often compress an entire high-resolution frame before decoding a coordinate string and label. LIR retains explicit multi-scale search, source-resolution evidence, adaptive compute, and calibrated abstention in one trainable model.

A. Presence-Aware Contextual Proposals

A shared encoder processes one global view and coordinate-aware adaptive tiles. A query bank contains an explicit null query and class-agnostic event queries. It predicts eventness

TABLE I

POSITIONING AGAINST REPRESENTATIVE AERIAL BENCHMARKS. “NO-EVENT” MEANS EXPLICIT ORDINARY/NEGATIVE FRAMES ARE PART OF THE TARGET EVALUATION; CONTEXT ROI DENOTES A SUFFICIENT-EVIDENCE REGION RATHER THAN A PIXEL-TIGHT OBJECT BOX. SHIFT REQUIRES GROUPED GEOGRAPHIC OR TEMPORAL TESTING.

Benchmark	Main scope	Real UAV	Region output	No-event	Context ROI	Full→event discovery	Site/time shift
ERA [3]	event recognition	✓	—	—	—	—	—
VisDrone [2]	object detection/tracking	✓	✓	—	—	—	—
A2Seek [14]	video anomaly reasoning	✓	✓	✓	—	✓	—
UAVBench [5]	image/region VL	✓	✓	—	—	—	—
AerialVG [9]	relational grounding	✓	✓	—	—	—	—
UniRef-UAV [11]	universal referring	✓	✓	✓	—	—	partial
MM-UAVBench [6]	perception–planning	✓	partial	—	—	—	—
SpatialUAV [7]	spatial intelligence	✓	✓	—	—	—	—
UTS-UAV	operational events	✓	✓	✓	✓	✓	✓



Fig. 2. Benchmark construction. The full contextual-box recovery and human acceptance are complete; no-event samples, labels, privacy clearance, company authorization, and sealed test evaluation are part of the frozen release.

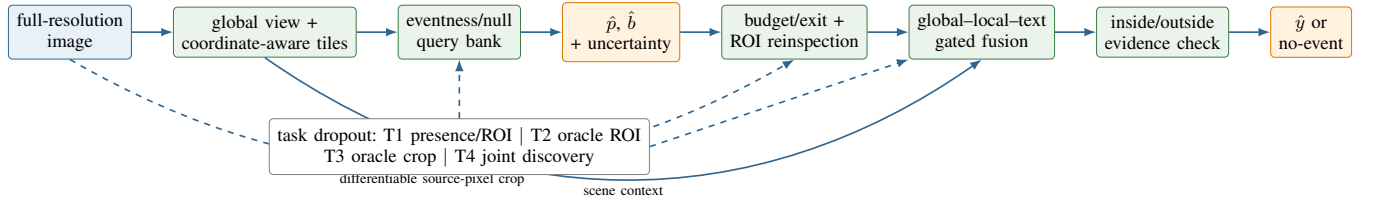


Fig. 3. LIR performs coarse-to-fine event discovery. A null/eventness query allows abstention on true negatives; positive hypotheses are reinspected from original pixels before local, global, and class-text evidence is fused and counterfactually verified. One parameter set supports T1–T4; T5 exposes the same heads as a cascade.

TABLE II

FIVE LINKED TASKS. POS.: POSITIVE EVENT FRAME; NEG.: NO-EVENT FRAME.

Task	Input	Output	Pos.	Neg.
T1	full frame	no-event or ROI	✓	✓
T2	full + GT ROI	event class	✓	—
T3	GT crop	event class	✓	—
T4	full frame	no-event or ROI+event	✓	✓
T5	full→pred. crop	no-event or ROI+event	✓	✓

$\hat{p}$, normalized contextual box $\hat{b}$, and uncertainty u_b . The class-agnostic head lets rare event classes share region supervision. Localization losses are gated by positive presence:

$$\mathcal{L}_{\text{search}} = \lambda_p \mathcal{L}_{\text{focal}}(p, \hat{p}) + p \left(\lambda_1 \|b - \hat{b}\|_1 + \lambda_g \mathcal{L}_{\text{GIoU}}(b, \hat{b}) \right). \quad (2)$$

Training batches balance ordinary negatives, visually similar hard negatives, and long-tailed positives. Online hard-negative mining selects high-eventness false alarms without using the test set.

B. Difficulty-Adaptive Anytime Search

After each nested inspection, a controller combines presence/class entropy, box uncertainty, and the inside–outside

evidence margin into residual risk. It exits under a validation-frozen risk constraint or selects the next tile or source-pixel proposal. A cost-regularized loss imitates the least expensive correct budget supplied by a training-only oracle. LIR and agent baselines use the same registered call ladder, and every inspection is charged.

C. High-Resolution Inspection and Semantic Fusion

For a positive hypothesis, LIR enlarges $\hat{b}$ by context ratio ρ chosen on validation and samples source pixels with differentiable ROI Align. ROI features z_r , global context z_g , and train-only class-text prototypes t_k are combined through a gated fusion head:

$$p(y = k | I, \hat{b}) = \text{softmax}_k (g_\theta [z_r, z_g, t_k, u_b]). \quad (3)$$

This preserves small evidence while preventing the ROI branch from discarding scene context. Class-balanced focal loss and logit adjustment address the long tail; temperature scaling on validation calibrates the final eventness and class confidence.

D. Counterfactual Evidence Verification

A geometrically plausible box may still be semantically incidental. For each candidate b_j , we form an inside view I_j^{in} from source pixels and an outside view I_j^{out} by replacing

the proposed region with a validation-frozen neutral texture. Let $s_y(\cdot)$ denote the pre-softmax score of the target event. We optimize

$$\mathcal{L}_{\text{ev}} = [m - s_y(I^{\text{in}}) + s_y(I^{\text{out}})]_+, \quad (4)$$

so an accepted ROI must preserve class evidence inside and remove it outside. At inference, a verifier reranks the top- K class-agnostic proposals by

$$q_j = \hat{p}_j p(\hat{y}_j | I, b_j) \sigma(s_{\hat{y}_j}(I_j^{\text{in}}) - s_{\hat{y}_j}(I_j^{\text{out}})). \quad (5)$$

If no candidate exceeds the validation-frozen selective threshold, the model abstains. The verifier uses the same features and source pixels as the base model; its FLOPs and both counterfactual passes are included in latency.

E. Multi-Task Curriculum and Inference

Task dropout alternates T1–T4. The semantic branch begins with oracle ROIs, then progressively mixes jittered and predicted ROIs. A symmetric KL term aligns class distributions from crop, oracle-box, and predicted-box views:

$$\begin{aligned} \mathcal{L} = & \mathcal{L}_{\text{search}} + p\lambda_c \mathcal{L}_{\text{cls}} + p\lambda_k D_{\text{SKL}}(p_{\text{oracle}}, p_{\text{pred}}) \\ & + p\lambda_e \mathcal{L}_{\text{ev}} + \lambda_v \mathcal{L}_{\text{view}}. \end{aligned} \quad (6)$$

At inference, verifier confidence below the validation-frozen threshold returns no-event; otherwise the model returns one canonical ROI–event record. Generative baselines use the same schema and invalid output is scored as failure. The unified and cascade comparisons account for every tile, crop, visual token, parameter, and millisecond.

V. EXPERIMENTS

All benchmark construction, contextual-box acceptance, governance, and test-label sealing are complete. Red cells reserve exact values for insertion from the frozen evaluator; red prose records the expected interpretation and must be checked against the completed runs before submission.

A. Models, Training, and Reproducibility

We evaluate five model families. **Crop/oracle classifiers** include ConvNeXt, ViT, DINOv2 [26], CLIP/SigLIP-style linear probes, Florence-2 [23], and Qwen3-VL [25]. **Closed-set localizers** include Faster R-CNN, RT-DETR, DINO, and D-FINE [22]. **Open-vocabulary grounders** include Grounding DINO [20] and YOLO-World [21], evaluated zero-shot and fine-tuned. We also initialize class-agnostic search from AerialVG, DVGBench, UniRef-UAV, and UAV open-vocabulary pretraining [9]–[11], [28]. **Generative systems** use Florence-2 and Qwen3-VL with a canonical no-event/box/event schema. **End-to-end systems** include every validation-selected localizer–classifier cascade, a Qwen3-VL search–inspect–verify tool agent on the registered call ladder, and LIR. A stratified human study supplies an operational reference for presence, ROI sufficiency, and event class.

All methods use identical record IDs, ontology, and train-only class definitions. Fine-tuned comparisons use three declared seeds, class-balanced sampling, and validation-selected early stopping. We evaluate global inputs at 640,

TABLE III
ORACLE REPRESENTATION STUDY ON IDENTICAL POSITIVE RECORDS.
CTX. DENOTES A VALIDATION-SELECTED CONTEXT MARGIN.

Model	Positive input	Acc.	Macro-F1	Valid
ConvNeXt	GT crop (T3)	TBD	TBD	100
DINOv2	GT crop (T3)	TBD	TBD	100
Florence-2	GT crop (T3)	TBD	TBD	TBD
Qwen3-VL	GT crop (T3)	TBD	TBD	TBD
Qwen3-VL	full + drawn ROI	TBD	TBD	TBD
Qwen3-VL	full + coordinates	TBD	TBD	TBD
Qwen3-VL	masked background	TBD	TBD	TBD
Qwen3-VL	ROI + ctx.	TBD	TBD	TBD

1024, and 1536 pixels, fixed $2 \times 2/3 \times 3$ tiling, and each model’s native dynamic-resolution policy. Checkpoint revision, prompt, precision, optimizer, schedule, LoRA rank, context margin, seed, GPU, and parser are frozen in a run manifest. Results report mean $\pm$ std across seeds and 95% session-bootstrap intervals. Latency includes preprocessing, all tiles/crops, generation, and postprocessing at batch size one.

B. RQ1: How Much Does the Input Representation Hide?

T3 measures recognition with a supplied source-resolution crop. T2 retains the correct ROI but varies how it is represented: crop, drawn box, coordinates, masked background, or ROI plus surrounding context. The paired T3–T2 gap separates visual compression/background interference from localization error. *Projected trend—verify and replace after the frozen run: Source-resolution crops should provide the highest recognition ceiling; drawn-box and coordinate prompts should expose whether VLMs truly use region cues, while removing all background may hurt diffuse contextual events.*

C. RQ2: Can Models Reject Negatives and Ground Positives?

T1 is evaluated as presence-aware proposal generation, not positive-only box regression. We compare global, tiled, and coarse-to-fine inputs; closed-set and open-vocabulary pretraining; single- and multi-scale features; and calibration before and after temperature scaling. Precision–recall and FROC curves show positive recall versus false alarms per 1,000 no-event frames. Localization is stratified by ROI-area quartile, context diffuseness, border contact, and negative subtype. *Projected trend—verify and replace after the frozen run: Tiling should primarily improve small-ROI recall, while hard-negative mining and calibration should reduce false alarms; open-vocabulary zero-shot models are expected to over-trigger on noun-like distractors.*

D. RQ3: Unified Model or Modular Cascade?

T4 and T5 share presence, grounding, recognition, and efficiency metrics. The cascade exposes interpretable intermediate boxes and can pair the strongest specialists; a unified model shares context and amortizes computation. The matched-compute comparison caps input pixels, visual tokens, and

TABLE IV

PRIMARY SESSION-DISJOINT RESULTS. P-AP: EVENT-PRESENCE AP; N-FPR: FALSE-POSITIVE RATE ON NO-EVENT FRAMES; AP50: CONTEXTUAL LOCALIZATION AP; C-F1: POSITIVE SUPPORTED-CLASS MACRO-F1; G-MAP50: END-TO-END EVENT-AWARE GROUNDING AP. LATENCY INCLUDES ALL TILES, CROPS, DECODING, AND POSTPROCESSING.

System	Family/mode	P-AP↑	N-FPR↓	AP50↑	C-F1↑	G-mAP50↑	ms↓
D-FINE	closed-set T1	TBD	TBD	TBD	–	–	TBD
Grounding DINO	open-vocab. T1	TBD	TBD	TBD	–	–	TBD
YOLO-World	open-vocab. T1	TBD	TBD	TBD	–	–	TBD
Qwen3-VL	generative T4	TBD	TBD	TBD	TBD	TBD	TBD
Florence-2	generative T4	TBD	TBD	TBD	TBD	TBD	TBD
D-FINE + DINOv2	modular T5	TBD	TBD	TBD	TBD	TBD	TBD
Best loc. + Qwen3-VL	modular T5	TBD	TBD	TBD	TBD	TBD	TBD
Qwen3-VL tool agent	search/inspect/verify	TBD	TBD	TBD	TBD	TBD	TBD
LIR (ours)	unified T4	TBD	TBD	TBD	TBD	TBD	TBD
Human reference	stratified subset	TBD	TBD	TBD	TBD	TBD	–

TABLE V

FROZEN DEPLOYMENT-SHIFT SHORTLIST. RET.: G-MAP50 RETENTION RELATIVE TO THE SESSION-DISJOINT SCORE; ALL THRESHOLDS REMAIN FIXED.

System	Session G-mAP50	Unseen site G-mAP50 / Ret.	Future time G-mAP50 / Ret.
Best generative VLM	TBD	TBD/TBD	TBD/TBD
Best cascade	TBD	TBD/TBD	TBD/TBD
LIR	TBD	TBD/TBD	TBD/TBD
Human reference	TBD	TBD/TBD	TBD/TBD

wall-clock budget. Errors are decomposed into false alarm, miss, localization-only, classification-only, joint, and invalid-output outcomes. *Projected trend—verify and replace after the frozen run: The best cascade should set a strong modular baseline, while LIR is expected to gain on G-mAP and latency through joint training and source-pixel reinspection; generic VLMs may retain higher output-invalid and negative-frame false-positive rates.*

E. RQ4: Does Performance Transfer Across Place and Time?

The complete matrix is selected on session-disjoint validation. A frozen shortlist—best localizer, classifier, cascade, generative VLM, and LIR—is then run without test adaptation on unseen-site and forward-temporal protocols. We report absolute score, relative retention, and bootstrap interval. *Projected trend—verify and replace after the frozen run: Unseen-site should produce the largest degradation because scene layout and background priors change together; the crop oracle is expected to degrade less than full-frame systems, identifying search rather than semantics as the dominant transfer bottleneck.*

F. RQ5–RQ8: Calibration, Scale, Tail, and Format

Negative calibration (RQ5). Reliability diagrams, ECE, Brier score, N-FPR@95% recall, and selective-risk curves compare raw, temperature-scaled, and class-conditional thresholds. Hard-negative categories are reported rather than averaged away. **Model scale and resolution (RQ6).** We sweep model capacity, input pixels, tile overlap, and number of visual

TABLE VI

LIR ABLATION UNDER MATCHED COMPUTE. VALUES ARE ABSOLUTE SCORES; Δ COLUMNS ARE RELATIVE TO THE FULL MODEL.

Variant	G-mAP50	Δ N-FPR	Δ ms
Full LIR	TBD	–	–
– null/eventness query	TBD	TBD	TBD
– adaptive tiles	TBD	TBD	TBD
– source-pixel reinspection	TBD	TBD	TBD
– evidence verifier	TBD	TBD	TBD
– global context	TBD	TBD	TBD
– class-text prototypes	TBD	TBD	TBD
– hard-negative mining	TBD	TBD	TBD
– oracle→pred. curriculum	TBD	TBD	TBD
– cross-view consistency	TBD	TBD	TBD

tokens, plotting G-mAP against latency and memory to expose the Pareto frontier. **Long-tail and region geometry (RQ7).** Head/medium/tail classes, effective-number reweighting, logit adjustment, balanced sampling, and few-shot support curves are crossed with ROI size and diffuseness. **Prompt and output format (RQ8).** Zero-shot, few-shot, class-definition, and negative-aware prompts are tested with JSON, coordinate tokens, and multiple-choice reformulations. Parsing failures remain failures.

Projected trend—verify and replace after the frozen run: Calibration should improve operating-point reliability without changing ranking metrics; resolution gains should saturate beyond the point at which ROI pixels are recoverable; class balancing should benefit macro-F1 more than overall accuracy; multiple-choice reformulation may improve validity but should not remove the underlying grounding gap.

G. RQ9: Which LIR Components Matter?

Each ablation changes one component under identical compute and seed schedules. The null query and hard-negative mining target ordinary-frame rejection; adaptive tiles target small evidence; global context and class text target diffuse/rare events; the oracle-to-predicted curriculum and view consistency target error propagation. We additionally compare independent heads, shared backbone only, and full multi-task training. *Projected trend—verify and replace after*

the frozen run: Removing source-resolution reinspection or adaptive tiles should hurt small ROIs most; removing the null query/hard negatives should raise N-FPR; removing global context should disproportionately hurt diffuse event classes.

H. RQ10: Data Scale, Label Efficiency, and Robustness

Training uses stratified 10/25/50/75/100% subsets at fixed validation/test sets, plus 1/2/4/8/16-shot tail-class probes. Annotation robustness is tested with ROI jitter, context expansion/contraction, and source-crop versus human-accepted-box supervision. Three seeds are used at every scale. Finally, an image-random split is reported only as a leakage diagnostic, never as the main benchmark. *Projected trend—verify and replace after the frozen run: Performance should follow a sublinear scaling law; presence and localization are expected to need more acquisition diversity than crop recognition, while moderate ROI perturbation should be tolerated because boxes encode context rather than exact object boundaries.*

I. RQ11–RQ14: Evidence, Robustness, Transfer, and Budget

Evidence sufficiency (RQ11). For GT and predicted ROIs, we compare inside-only recognition, outside-only recognition, neutral-region deletion, context-preserving insertion, random boxes matched for area, and saliency boxes matched for compute. We report ESG, per-class confidence drop, and human sufficiency preference. The verifier is disabled at train time, test time, and both to separate supervision from reranking. **Physical robustness (RQ12).** Seven UAV-relevant corruptions—motion/defocus blur, JPEG, haze, under/overexposure, downsampling, and roll—are evaluated at five severities following corruption-benchmark practice [29]. G-mAP retention, N-FPR inflation, calibration drift, and consistency of the returned ROI are reported. **Public aerial transfer (RQ13).** ImageNet, COCO, DINOv2, aerial-object grounding, implicit grounding, and UAV open-vocabulary initializations are compared under zero-shot, linear-probe, LoRA, and full fine-tuning budgets. We additionally report Kendall rank correlation across session, site, time, and corruption protocols to test whether model selection itself transfers. **Difficulty-adaptive search (RQ14).** Global-only, fixed- K tiling, confidence-adaptive, learned, and training-only oracle routing are compared at 1/2/4/8/16 calls for both LIR and the Qwen3-VL agent. We report G-mAP/N-FPR curves, budget AUC, oracle regret, early-exit errors, and wall-clock cost.

Projected trend—verify and replace after the frozen run: Evidence verification should help diffuse classes; blur/downsampling should damage small-ROI search; aerial pretraining should accelerate localization but need contextual fine-tuning; adaptive search should save calls on easy negatives while spending them on compact/diffuse positives.

VI. BENCHMARK DESIGN AND IMPLICATIONS

Why a 0/1 contextual ROI formulation? Operational frames in the present release are naturally described by one primary decision target: either no benchmark event is visible, or one region contains the evidence used to assign the event

label. This formulation is more realistic than positive-only classification and more faithful than converting operator crops into pixel-tight object boxes. It also prevents an evaluation loophole in which a model always emits a plausible region. A useful method must first earn the right to localize by assigning sufficient confidence that an event is present. Multi-event and exhaustive multi-instance annotation remain valuable future extensions, but they answer a different operational question.

How should contextual grounding be interpreted? A contextual ROI is not an imprecise substitute for segmentation. Its boundary defines an evidence budget: the selected pixels should be sufficient for a trained reviewer to judge the event while excluding irrelevant scene area. Consequently, no single IoU threshold is adequate. AP/R at 0.25, 0.50, and 0.75, normalized center error, ROI-area slices, and jitter robustness jointly measure whether a model searches the correct neighborhood and whether its semantic head tolerates reasonable boundary variation. The T2/T3 pairing is equally important: if crop recognition is strong but oracle-box recognition is weak, compression or prompting is the bottleneck; if both are strong while T4 is weak, search or calibration is responsible.

IoU and center error still cannot prove that the selected pixels caused the decision. ESG therefore asks a complementary question: does retaining the ROI preserve recognition, and does removing it destroy recognition? Random area-matched and saliency-matched controls prevent a large-box or generic attention advantage. ESG remains diagnostic rather than a leaderboard target because it depends on frozen recognition probes; agreement across two probes and the human sufficiency subset is reported explicitly.

Leaderboard organization. We recommend three non-collapsible tracks. The *Oracle Recognition* track reports T2/T3 macro-F1 and tail behavior; the *Presence and Grounding* track reports T1 P-AP, N-FPR, contextual AP, and calibration; the *End-to-End Discovery* track reports T4/T5 G-mAP, valid-output rate, and latency. A single overall average is discouraged because it can hide unsafe false-alarm behavior or trade a severe miss rate for easy crop accuracy. Instead, the leaderboard exposes a Pareto view of G-mAP, N-FPR, latency, and memory, and requires results on all three deployment protocols. Test-time adaptation, external training data, model size, input-pixel budget, and manual output repair must be declared as separate leaderboard metadata.

The supplement fixes closed/external-data, parameter-efficient, and agentic submission modes, sealed-server receipts, and the anytime challenge curve.

Implications for operational deployment. The benchmark is designed for decision support rather than autonomous enforcement. Different inspection programs incur different costs for misses and false alarms, so validation-frozen operating points and selective-risk curves are more actionable than accuracy alone. Calibrated abstention can route uncertain frames to human review, while the predicted contextual ROI supplies an interpretable evidence view. Site/time retention and hard-negative slices should determine whether a model is eligible for a new route; an aggregate session score is

insufficient. These reporting rules turn benchmark progress into measurable improvements in search reliability, review burden, and geographic transfer rather than merely higher crop-classification accuracy.

VII. LIMITATIONS, ETHICS, AND INTENDED USE

Each positive record supplies one primary contextual ROI; the benchmark does not measure exhaustive instances or segmentation. Diffuse ROI edges motivate multi-threshold, center-error, and jitter analyses instead of a single strict IoU. The ontology is region-informed, rare classes remain difficult, unseen sites do not imply global coverage, and images do not evaluate tracking, causal explanation, or autonomous action.

The data support perception research and decision support, not autonomous enforcement. Authorization, privacy/figure review, and test-label sealing are complete. Public artifacts use opaque site/session IDs and exclude precise routes, coordinates, and unnecessary identifiers; protected test annotations remain private. Models may encode geographic or socioeconomic bias, so results must be read by site, time, class, and false-alarm slice rather than as a universal safety claim.

VIII. CONCLUSION

UTS-UAV turns operational UAV event discovery into an auditable chain of no-event rejection, contextual grounding, and recognition. Five tasks, three deployment shifts, broad baselines, and fourteen diagnostics expose each bottleneck. LIR unifies source-pixel reinspection, global–local–text fusion, counterfactual evidence verification, and adaptive compute. Registered matched-budget, corruption, and aerial-transfer tests determine whether gains survive beyond a favorable representation. *Result placeholder: Replace with the final quantitative conclusion: strongest gain, budget–accuracy AUC gain, evidence-sufficiency gain, remaining human gap, dominant failure mode, and cross-shift retention.*

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